

Adaptive Fabric Surface Defect Detection and Localization Using Classical Texture Descriptors

Team TEXTURA

Weekly Project Progress Report – Week 2

Course: Digital Image Processing

Repository link:

https://github.com/AditiKapadiya/DIP_2026_Project17_Group8.git

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Work done in Week 2

Week 2 was aimed at cleaning the raw AITEX images and masks and transforming them into a clean, validated, model-ready dataset, building on the dataset understanding from Week 1. The work is broken down to four stages namely: integrity checking, preprocessing, patch generation, and validation.

1. Dataset Integrity Checks

All defect images were verified prior to processing to ensure that they will ultimately be usable as the dataset. When processing 106 defect images and 107 mask files:

There were 103 images that matched perfectly with 1 mask.

1 image (0100_025_08.png) had no matching mask at all.

Each of these two images (0044_019_04.png and 0097_030_03.png) has more than one defect region and thus two separate masks.

1 mask (0106_010_03_mask.png) was empty, that is, a file without any defect pixels.

The dimensions of the image and mask were verified in each valid pair, and there was no mismatch in the size of the image and mask in the source.

These results were compiled into a list of issues, and then the images remaining for future selection were determined according to these issues.

2. Preprocessing

A preprocessing step was applied to the images which passed the integrity check:

The normalized contrast was done in each image by applying the CLAHE (Contrast Limited Adaptive Histogram Equalization) technique to the luminance channel of each image, which was used to normalize the contrast between different parts of the fabric surface.

After the contrast enhancement, a light Gaussian blur was performed to decrease the noise caused by the contrast enhancement.

A logical OR was used to combine images and masks with multiple defect regions into a single binary mask, thus each image will have only one mask in the following.

Images and masks were resized to 256×256 to do a first-pass check of pipeline (105 images processed successfully at this point), and the masks were checked for only binary values after resizing (0, 255).

3. Patch Generation

The size of the defects can be small relative to the image size, and so the images were broken into patches of fixed size (256×256) using a sliding window with 128-pixel stride (50% overlap) in addition to the resized full images:

The source for the defect patches, after discarding defective images with missing-mask and empty-mask (as mentioned above) was 104 usable defective images.

This resulted in a total of 3,222 patches of which 367 contain defect pixels and 2855 contain only background pixels.

The defect-free images were subjected to the same CLAHE and denoising preprocessing steps, resulting in 4371 more background patches for the defect-free class.

Each file name was used to extract fabric type for fabric-wise analysis later on in the week.

4. Validation

The patch dataset was validated, end-to-end:

All image patches were verified as having a corresponding mask patch; no image patches were found without a corresponding mask patch.

It was checked that all the patches (images and masks) are exactly 256×256.

All patch masks were verified as having only binary values (0 and 255), leaving no "in between" pixel values from resizing or OR-combination step.

As a final sanity check before proceeding, a visual check was added at the end, which showed one of the defect patches along with the corresponding mask.

Literature Review

In the textile manufacturing industry, automated fabric-defect detection is a crucial computer vision application because it is prone to time constraints, fatigue, and inconsistency when performed by humans. Hence, the study of past research has investigated image-processing and deep-learning techniques to detect and locate defect in fabric images (Rasheed et al., 2020).

One of the most popular datasets for this problem is the AITEX Fabric Image Database (AFID). It has 245 images of 7 types of fabric, 140 of which are defect-free and 105 are defective. The resolution of original images is 4096×256 and the defective images have corresponding pixel level masks, with a white pixel marking a defect area. The dataset consists of 12 classes of defects, thus it can be used for defect detection and segmentation (AITEX,2025).

Previous works have demonstrated that the size of the images to be processed in AITEX is sufficiently large to be helpful for processing with patch-based approaches. In an example, the original 4096×256 images have been divided into 256×256 sub-images, to increase the number

of training samples and increase the easier segmentation of defects.(Xu et al.,2023) This conforms to our Week 2 strategy of creating fixed size patches first, then training the models.

The image processing is also crucial since illumination change, noise and regions of low contrast in fabric images may exist. Recent studies on fabric defects pre-processing using OpenCV processing have found that the use of Gaussian filtering and CLAHE to remove noise and enhance the contrast of the defect region before the application of defect models can reduce the effects of noise. (Xu et al.,2026) In our implementation, we enhance the contrast of the luminance channel of LAB colour space, and use Gaussian denoising to improve the image quality.

As a result, the preprocessing pipeline used in Week 2 is aligned with the current literature, with each step in the processing of the data being described: dataset validation, mask preparation, contrast enhancement and denoising, patch generation and patch level validation. This will give the following machine-learning/deep-learning steps a uniform dataset.

Challenges Faced

1. Image–mask pairing

The biggest problem was to be able to make sure that each defective image was matched with the proper mask. The AITEX naming convention includes image identifiers, defect codes and fabric codes and assuming that ordering the images correctly would result in incorrect code pairing. Thus, a mask lookup by image ID and explicit checks for the nonexistence or multiplicity of masks have been developed.

2. Multiple masks for single image

The design of a mask depends on the image being created. There may be multiple masks for some images. These are not the only images that are processed; the masks are joined together using a logical OR, which allows all defect areas to be captured in a single binary mask.

3. Invalid/Empty masks

The other challenge was to find masks that were technically available but had no defect pixels. It was necessary to detect these cases individually as a mere check of if a mask file exists was not enough.

4. Maintaining image–mask alignment

Different interpolation methods are needed for resizing the images and masks. The notebook interpolates images using bilinear, and masks using nearest neighbour, interpolation. This is critical as it can cause unwanted gray values to be interpolated into binary masks..

5. Original image dimension handling

The AITEX images are very wide (4096×256). To make fixed-size patches, one had to be mindful of the edges of the image. The patch-generation function has a stride of 128 pixels and a patch size of 256×256 , with an overlap of 128 pixels, and explicitly specifies that the last row and column are covered.

6. Ensuring the quality of binary masks.

The masks were then checked for being either 0 or 255 after the preprocessing and patch extraction. This validation was needed to avoid misleading of the pre-processing steps to corrupt the segmentation labels.

7. Dataset imbalance

Another challenge is that most of the pixels of fabric images correspond to normal fabrics instead of defects. Thus, there are several background-only patches that can be obtained in patch extraction than defect-containing patches. The notebook, therefore, counts the number of defects containing patches and background-only patches separately, and also creates patches from images without defects.

8. Computational and storage applications.

The more high resolution pictures that are processed and the more patches that overlap, the more time will be consumed in processing and the more space will be used. The stride is 128 pixels, so there are more samples generated due to the overlap of neighbouring patches.

Tasks to be done in week 3:

LBP-Based Baseline Model

In Week 3, attention will be paid to creating the first baseline defect detection model using Local Binary Patterns (LBP). Preprocessed fabric patches will be used for the implementation of LBP, which will help in capturing local texture patterns and highlight the difference between patches due to defects. To find out appropriate parameters for the dataset, different LBP parameters including the radius and the number of neighboring pixels will be tested.

Normal texture characteristics will be created using defect-free patches for each type of fabric. These normal texture patterns will then be compared to the LBP histogram of test patches with a proper distance measure, for example chi-square distance, to obtain a

score called texture deviation or anomaly score. The higher the number of deviations, the more likely that there will be a defect.

Initial binary defect masks will be created from the anomaly maps with an appropriate threshold. A visual comparison will also be made using the masks predicted by the model to the original fabric images. Evaluation will be done with basic metrics, namely IoU, precision and recall, on a validation subset.

Deliverable: A working LBP defect detection pipeline that includes initial defect maps, visualizations and baseline performance results.

References

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